ORBITAL MECHANICS OF PLANETS — GRADUATE EXAM

Total Points: 50.0 | Questions: 5 | Date: March 31, 2026

Diagram-based questions on Orbital Mechanics of Planets

Question 1 10.0 points

Examine the diagram below depicting a planet's elliptical orbit around the sun. Using Kepler's second law, describe how the area swept out by a line from the planet to the sun varies over time. Explain how the speed of the planet changes as it moves along different segments of its orbit and why.

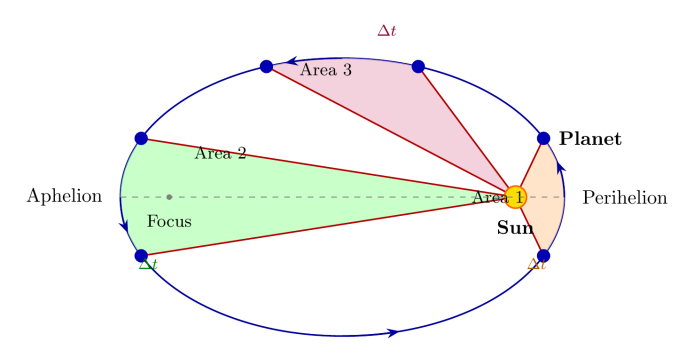

Fig. 1. An elliptical orbit of a planet around the sun. The sun is located at one of the foci of the ellipse. The orbit should be divided into several segments, each representing equal time intervals. Arrows should indicate the direction of the planet's motion. The areas swept out by the line from the planet to the sun over these intervals should be shaded or marked to illustrate Kepler's second law, which states that these areas are equal. Additionally, indicate the varying speed of the planet along different segments of the orbit, with faster motion near the perihelion and slower motion near the aphelion.

Question 2 10.0 points

Refer to the diagram below showing a spacecraft executing a Hohmann transfer orbit between two circular orbits around Earth. Calculate the velocity changes (Δv) required at each segment of the transfer. Provide a detailed step-by-step explanation of the calculation.

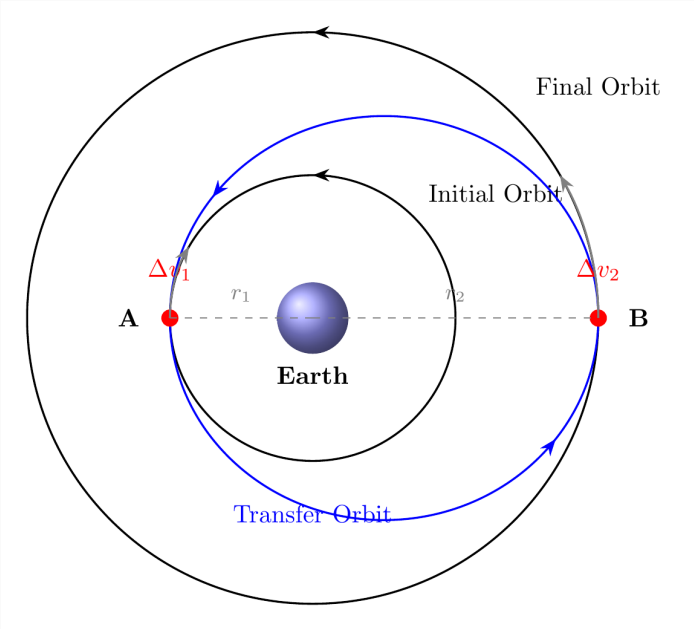

Fig. 2. REQUIRED LABELS (must appear visibly in the diagram): Δv A diagram showing a spacecraft executing a Hohmann transfer orbit between two circular orbits around Earth. The diagram includes: - Two concentric circles representing the initial and final orbits around Earth. - An elliptical path representing the transfer orbit connecting the two circular orbits. - Labeled points indicating the positions where velocity changes (Δv) occur: at the perigee of the transfer orbit (where it intersects the initial orbit) and at the apogee of the transfer orbit (where it intersects the final orbit). - Earth at the center of the orbits. IMPORTANT: The diagram is missing required labels. The diagram is missing the required label "Δv. Ensure ALL structural components mentioned in the description are clearly labeled. IMPORTANT: The diagram is missing required labels. The diagram description explicitly states that the label ". Ensure ALL structural components mentioned in the description are clearly labeled.

Question 3

10.0 points

Analyze the diagram below illustrating the trajectory of a space probe performing a gravitational slingshot maneuver around a planet. Explain how the slingshot alters the probe's trajectory and velocity. Discuss the conservation laws applicable in this maneuver, considering <eq_q_1> and <eq_q_2>.

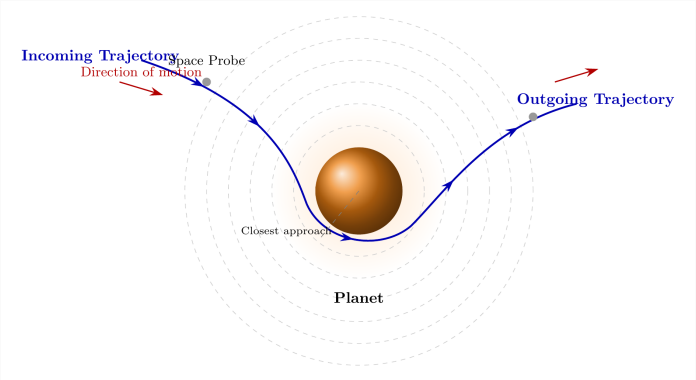

Fig. 3. A diagram illustrating the trajectory of a space probe performing a gravitational slingshot maneuver around a planet. The diagram should show the planet at the center with its gravitational field indicated by concentric circles. The space probe's trajectory should be depicted as a curved path approaching the planet, looping around it, and then departing in a different direction. Arrows should indicate the direction of the probe's motion before and after the maneuver. Include labels for the probe's velocity vector before and after the maneuver, and highlight the change in trajectory. The planet should be labeled, and the probe's path should be clearly marked to show the slingshot effect.

Question 4

10.0 points

Examine the diagram below that illustrates the five Lagrange points (L1-L5) in the Earth-Sun system. Describe the stability of each point and how satellites can utilize these positions.

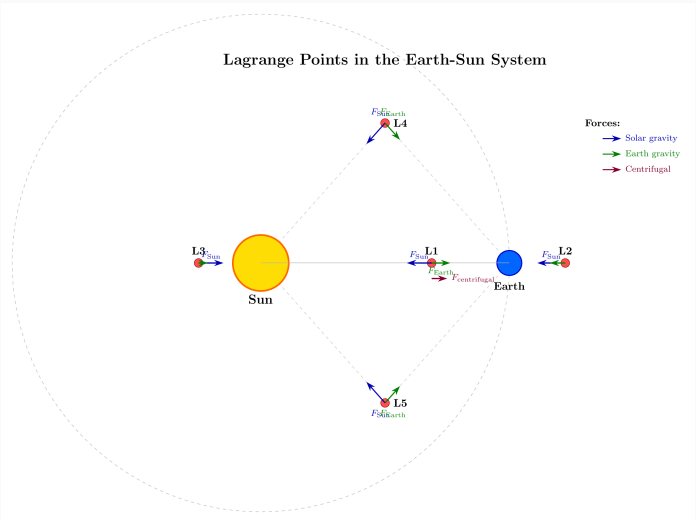

Fig. 4. A diagram illustrating the five Lagrange points (L1-L5) in the Earth-Sun system. The Sun and Earth are shown, with the Lagrange points labeled and positioned relative to the Earth and Sun. L1 is between the Earth and Sun, L2 is on the opposite side of the Earth from the Sun, L3 is on the opposite side of the Sun from the Earth, L4 and L5 form an equilateral triangle with the Earth and Sun. Include arrows to indicate the direction of gravitational forces and label the points clearly.

Question 5 10.0 points

Referencing the diagram below that details Mercury's orbit, explain the phenomenon of orbital precession and its causes, including both Newtonian mechanics and general relativity contributions, with equations <eq_q_1> and <eq_q_2> resolved.

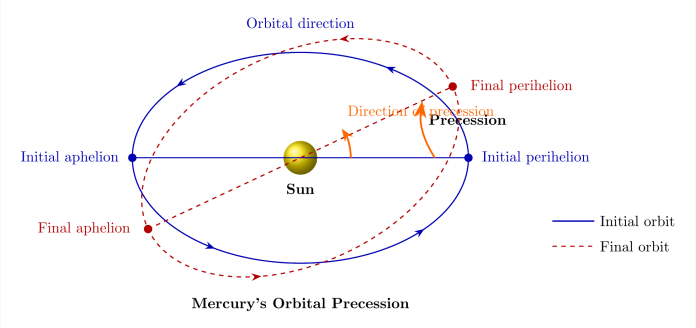

Fig. 5. Diagram of Mercury's orbit showing the elliptical path around the Sun, with an indication of the precession of the perihelion. The diagram should include the Sun at one focus of the ellipse, the perihelion point, and arrows indicating the direction of precession. The orbit should be labeled, and the precession angle should be marked to illustrate the shift over time.